

Unit 5 Linear Programming

Linear programming is a technique to obtain the solution of a problem containing the constraints under which the problem is to be solved.

This LP contains non-negative decision variables.

A general LP contains

- (i) An objective function (maximisation or minimisation)
- (ii) Constraints
- (iii) Non-negative decision variables

There are 3 methods to solve LPP

- (i) Graphical method
- (ii) Simple method
- (iii) Two-phase simple method

Ex 1. Graphical method:

Solve the following problem using graphical method
Maximize $Z = 3x + 4y$. Subjected to constraints
 $2x + y \leq 40$, then $2x + 5y \leq 180$, $x, y \geq 0$. Solve by graphical method.

$$Z = 3x + 4y \rightarrow \text{objective function}$$

$$2x + y \leq 40, 2x + 5y \leq 180 \rightarrow \text{constraints}$$

$$x, y \geq 0 \rightarrow \text{non-negative decision variables}$$

$$(i) 2x + y = 40$$

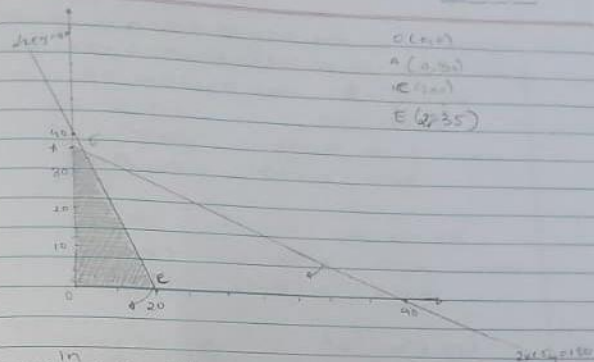
$$\text{put } y = 0 \Rightarrow x = 20 \rightarrow (20, 0)$$

$$x = 0 \Rightarrow y = 40 \rightarrow (0, 40)$$

$$(ii) 2x + 5y = 180$$

$$\text{put } y = 0 \Rightarrow x = 90 \rightarrow (90, 0)$$

$$x = 0 \Rightarrow 36 = y \rightarrow (0, 36)$$



* Region which all the constraints are satisfied is called Feasible region.

$$FR \rightarrow O(0,0) \quad A(0,36) \quad E(25,35) \quad C(20,0)$$

$$O(0,0) \rightarrow Z = 0$$

$$C(20,0) \rightarrow Z = 60$$

$$A(0,36) \Rightarrow Z = 144$$

$$E(25,35) \Rightarrow Z = 147.5 \rightarrow \text{Max}$$

Thus,

* Feasible Solution:

This is a solution which is a set of values for the decision variable that satisfies all the constraints & non-negative restriction simultaneously.

* Optimal Solution:

This is a feasible solution that achieves best possible value to the objective function.

Q. Using Graphical Method minimize $Z = 80x_1 + 10x_2$,
subjected to the constraint $x_1 + 2x_2 \leq 40$,
 $3x_1 + x_2 \geq 30$, then $4x_1 + 3x_2 \geq 60$, $x_1, x_2 \geq 0$

$$x_1 + 2x_2 = 40$$

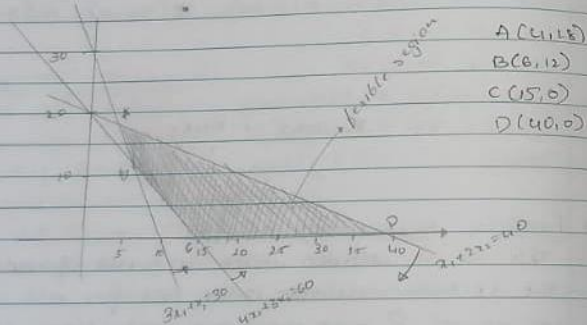
when $x_1 = 0$ $x_2 = 20$ (0, 20)
 $x_2 = 0$ $x_1 = 40$ (40, 0)

$$3x_1 + x_2 = 30$$

when $x_1 = 0$ $x_2 = 30$ (0, 30)
 $x_2 = 0$ $x_1 = 10$ (10, 0)

$$4x_1 + 3x_2 = 60$$

when $x_1 = 0$ $x_2 = 20$ (0, 20)
 $x_2 = 0$ $x_1 = 15$ (15, 0)



$$A(4, 16) \Rightarrow Z = 860$$

$$B(6, 12) \Rightarrow Z = 240 \rightarrow \text{min (OS)}$$

$$C(15, 0) \Rightarrow Z = 300$$

$$D(40, 0) \Rightarrow Z = 800 \rightarrow \text{max}$$

B(6, 12) is optimal solution.

Q. Minimize $Z = 80x + 16y$ Constraints: $6x + 2y \geq 12$,
 $2x + 2y \geq 8$, $4x + 12y \geq 20$ Find out feasible region
in clear.

$$6x + 2y = 12$$

when $x = 0$ $y = 6$ (0, 6)
 $y = 0$ $x = 2$ (2, 0)

$$2x + 2y = 8$$

when $x = 0$ $y = 4$ (0, 4)
 $y = 0$ $x = 4$ (4, 0)

$$4x + 12y = 20$$

when $x = 0$ $y = 5/3 = 1.66$
 $y = 0$ $x = 5$

* BIG-M method * (method of penalties) Page-(90)

STEP-①: Express the problem in standard form

STEP-②: Add non-negative variable [ARTIFICIAL VARIABLE $A_1, A_2, A_3, \dots$] to the left hand side of all those constraints which are of $\geq$ or $=$ type.

STEP-③: Assign a very large PENALTY ($-M$) to these artificial variables in the objective function.

STEP-④: solve the modified LPP by SIMPLEX METHOD

NOTE: Once an artificial variable leaves the basis, the column for this variable is **OMITTED** from the next simplex table

At any iteration of simplex method, one of the following three cases may arise

Sl. No.	POSSIBILITIES	CONCLUSION
1	NO ARTIFICIAL VARIABLE in the BASIS + OPTIMALITY CONDITION is satisfied	Solution is OPTIMAL BASIC FEASIBLE SOLUTION
2	There is at least ONE artificial variable in BASIS at ZERO LEVEL with zero value in b column + OPTIMALITY CONDITION is satisfied	Solution is DEGENERATE OPTIMAL BASIC FEASIBLE SOLUTION
3	There is at least ONE artificial variable in BASIS at NON ZERO LEVEL with positive value in b column + OPTIMALITY CONDITION is satisfied	Solution is NO FEASIBLE SOLUTION

Q) solve the following LPP by BIG-M method

$$\text{minimize } Z = 2x_1 + x_2$$

subject to:

$$3x_1 + x_2 = 3$$

$$4x_1 + 3x_2 \geq 6$$

$$x_1 + 2x_2 \leq 3$$

$$x_1, x_2 \geq 0$$

First convert it to MAXIMIZATION problem

$$\text{max } Z' = -2x_1 - x_2$$

Here Z' is $(-Z)$

Soln:- consider the first constraint:

$$3x_1 + x_2 = 3$$

(= Type)

∴ We have to add
ARTIFICIAL VARIABLE

$$3x_1 + x_2 + A_1 = 3$$

consider the second constraint:

$$4x_1 + 3x_2 \geq 6$$

(≥ Type)

Introduce SURPLUS VARIABLE S_1

+

Add Artificial variable A_2

∴

$$4x_1 + 3x_2 - S_1 + A_2 = 6$$

consider the third constraint:

$$x_1 + 2x_2 \leq 3$$

We have to just introduce
SLACK VARIABLE s_2 here

$$\therefore x_1 + 2x_2 + s_2 = 3$$

Thus, modified LPP is

$$\text{MAX } Z' = -2x_1 - x_2 + 0s_1 + 0s_2 - MA_1 - MA_2$$

$$\begin{aligned} \text{subject to: } & 3x_1 + x_2 + 0s_1 + 0s_2 + A_1 + 0A_2 = 3 \\ & 4x_1 + 3x_2 - s_1 + 0s_2 + 0A_1 + A_2 = 6 \\ & x_1 + 2x_2 + 0s_1 + s_2 + 0A_1 + 0A_2 = 3 \end{aligned}$$

$$x_1, x_2, s_1, s_2, A_1, A_2 \geq 0$$

INITIAL SIMPLEX TABLE:

Page-(25)

Surplus variable S_1 is not a basic variable.

Because, when initially $x_1=0$ & $x_2=0$ (Non-basic),

from second constraint, we get $S_1 = -6$.

As negative quantities are not feasible, S_1 must be prevented from appearing in the initial solution.

This is done by taking $S_1 = 0$

BASIC
VARIABLE

∴ Initial basic feasible solution is:

$$x_1=0, x_2=0, S_1=0$$

Non-basic variable

$$A_1=3, A_2=6, S_2=3$$

↓
from 1st
condition

↓
from 2nd
condition

↓
from 3rd
condition

∴ The initial simplex table is:

C_B	C_j BASIS	-2 x_1	-1 x_2	0 s_1	0 s_2	-M A_1	-M A_2	b	θ
-M	A_1	3	1	0	0	1	0	3	$3/3 = 1$
-M	A_2	4	3	-1	0	0	1	6	$6/4 = 1.5$
0	s_2	1	2	0	1	0	0	3	$3/1 = 3$
$Z_j = \sum C_B x_j$		-7M	-4M	M	0	-M	-M	-9M	
$C_j - Z_j$		7M-2	4M-1	-M	0	0	0		

← MINIMUM POSITIVE RATIO

∴ outgoing variable is

A_1

Delete A_1 column in the next table

↑ LARGEST POSITIVE VALUE (∵ M is LARGE) ∴ Incoming variable is x_1

Iteration (01):

Page-(27)

Introduce x_1 & drop A_1 from the BASIS.

Also, Divide ROW(1) by the key element 3

C_B	C_j BASIS	-2 x_1	-1 x_2	0 s_1	0 s_2	-M A_2	b	θ
-2	x_1	1	$\frac{1}{3}$	0	0	0	1	
-M	A_2							
0	s_2							
Z_j								
$C_j - Z_j$								

Now convert all the elements of **KEY COLUMN**

Page-(28)

to zero using elementary row operation.

$$R_2 \rightarrow R_2 - 4(\text{NEW } R_1)$$

$$R_3 \rightarrow R_3 - (\text{NEW } R_1)$$

C_B	C_j BASIS	-2 x_1	-1 x_2	0 s_1	0 s_2	-M A_2	b	θ
-2	x_1	1	1/3	0	0	0	1	$1/(1/3) = 3$
-M	A_2	0	5/3	-1	0	1	2	$\frac{2}{(5/3)} = \frac{6}{5}$
0	s_2	0	5/3	0	1	0	2	$\frac{2}{(5/3)} = \frac{6}{5}$
Z_j		-2	$-\frac{2}{3} + \frac{5M}{3}$	M	0	-M	-2-2M	
$C_j - Z_j$		0	$\frac{5M}{3} - \frac{1}{3}$	-M	0	0		

↑ LARGEST +ve value
∴ INCOMING VARIABLE is x_2

∴ outgoing variable is A_2
delete A_2 column from next table

← minimum POSITIVE RATIO

Iteration - (02):

Page - (29)

Introduce x_2 and drop A_2 from the basis

Also, divide ROW - (2) by the key element $5/3$

C_B	C_j BASIS	-2 x_1	-1 x_2	0 s_1	0 s_2	b	θ
-2	x_1						
-1	x_2	0	1	$-\frac{3}{5}$	0	$\frac{6}{5}$	
0	s_2						

NOW, convert all the elements of **KEY COLUMN** to ZERO using elementary row operation

$$R_1 \rightarrow R_1 - \left(\frac{1}{3}\right) \text{NEW } R_2, \quad R_3 \rightarrow R_3 - \left(\frac{5}{3}\right) \text{NEW } R_2$$

C_B	C_j BASIS	-2 x_1	-1 x_2	0 s_1	0 s_2	b	θ
-2	x_1	1	0	1/5	0	3/5	
-1	x_2	0	1	-3/5	0	6/5	
0	s_2	0	0	1	1	0	
Z_j		-2	-1	1/5	0	$-\frac{12}{5}$	
$C_j - Z_j$		0	0	-1/5	0		

Observe that none of $C_j - Z_j$ value
is positive

Page - (31)

∴ This is OPTIMAL SOLUTION

From the BASIS column & b column, we have

BASIS	b
x_1	$3/5$
x_2	$6/5$
s_2	0
Z_j	$-12/5$

∴ $x_1 = \frac{3}{5}$, $x_2 = \frac{6}{5}$

$\max Z' = -\frac{12}{5}$

But, the given problem is
of minimization type

∴ $\min(Z) = -\max(Z') = \frac{12}{5}$

PROBLEMS ON SIMPLEX METHOD

Q) solve the following LPP by SIMPLEX METHOD

minimize $Z = x_1 - 3x_2 + 3x_3$, subject to

$$3x_1 - x_2 + 2x_3 \leq 7,$$

$$2x_1 + 4x_2 \geq -12$$

$$-4x_1 + 3x_2 + 8x_3 \leq 10$$

$$x_1, x_2, x_3 \geq 0$$

convert the given problem to Standard form

condition ①: Objective function must be of MAXIMIZATION type

condition ②: All constraints should be expressed as EQUATIONS

condition ③: RHS of each constraint has to be non-negative

condition ④: All the variables $(x_1, x_2, \dots, x_n)$ must be non-negative

Soln:-

condition-①

Page-②

Given problem is MINIMIZING the objective function

$$Z = x_1 - 3x_2 + 3x_3.$$

converting it to maximization form, we set the objective function as

$$\max Z' = \min(Z)$$

↓

$$Z' = -Z$$

$$\therefore \max Z' = -x_1 + 3x_2 - 3x_3$$

min Z
is equivalent to
 $\max (-Z)$

Let us check condition - (3) first before (2),

page - (3)

condition - (3)

RHS of each constraint must be non-negative

From the second constraint, we have

$$2x_1 + 4x_2 \geq -12$$

-ve

multiply by (-1),

$$-2x_1 - 4x_2 \leq 12$$

condition-②

All the constraints must be expressed as EQUATIONS

First constraint:

$$3x_1 - x_2 + 2x_3 \leq 17$$

This is equivalent to

$$3x_1 - x_2 + 2x_3 + s_1 = 17$$

Here $s_1 \geq 0$ is known as

SLACK VARIABLE

second constraint:

$$-2x_1 - 4x_2 \leq 12$$

This is equivalent to

$$-2x_1 - 4x_2 + s_2 = 12$$

↓
 $s_2 \geq 0$ is slack variable

Third constraint:

$$-4x_1 + 3x_2 + 8x_3 \leq 10$$

This is equivalent to

$$-4x_1 + 3x_2 + 8x_3 + s_3 = 10$$

↓
 $s_3 \geq 0$ is slack variable

condition-④

All the variables must be non-negative.

This condition is already satisfied.

∴ given that $x_1 \geq 0$, $x_2 \geq 0$ & $x_3 \geq 0$

∴ Given linear programming problem is

$$\max Z' = -x_1 + 3x_2 - 3x_3 + 0s_1 + 0s_2 + 0s_3$$

subject to:

$$3x_1 - x_2 + 2x_3 + s_1 + 0s_2 + 0s_3 = 7 \quad \text{---①}$$

$$-2x_1 - 4x_2 + 0x_3 + 0s_1 + s_2 + 0s_3 = 12 \quad \text{---②}$$

$$-4x_1 + 3x_2 + 8x_3 + 0s_1 + 0s_2 + s_3 = 10 \quad \text{---③}$$

$$x_1, x_2, x_3, s_1, s_2, s_3 \geq 0$$

Initially set $x_1 = x_2 = x_3 = 0$ (non-basic solution) Page-(7)

When $x_1 = x_2 = x_3 = 0$, we get

From Eqn ①: $s_1 = 7$

From Eqn ②: $s_3 = 10$

From Eqn ③: $s_2 = 12$

BASIC SOLUTION

consider the following table:

CHECK THE OBJECTIVE FUNCTION
 $z' = -x_1 + 3x_2 - 3x_3 + 0s_1 + 0s_2 + 0s_3$

C_B	C_j basis	-1 x_1	3 x_2	-3 x_3	0 s_1	0 s_2	0 s_3	b	θ
0	s_1								
0	s_2								
0	s_3								
$Z_j = \sum C_B a_{ij}$									
$G_j = C_j - Z_j$									

From constraints ①, ② & ③, we have

page-(8)

$$A = \begin{bmatrix} 3 & -1 & 2 & 1 & 0 & 0 \\ -2 & -4 & 0 & 0 & 1 & 0 \\ -4 & 3 & 8 & 0 & 0 & 1 \end{bmatrix}$$

$$b = \begin{bmatrix} 7 \\ 12 \\ 10 \end{bmatrix}$$

Incorporate these rows in the table

C_B	C_j BASIS	-1 x_1	3 x_2	-3 x_3	0 s_1	0 s_2	0 s_3	b	θ
0	s_1	3	-1	2	1	0	0	7	
0	s_2	-2	-4	0	0	1	0	12	
0	s_3	-4	3	8	0	0	1	10	
$Z_j = \sum C_B a_{ij}$									
$C_j = c_j - Z_j$									

NOW, construct the row $Z_j = \sum C_B a_{ij}$

Example
1st element: $(0)(3) + 0(-2) + 0(-4) = 0$

$\downarrow$
 C_B $\downarrow$
 Find
 column

NEXT, construct the row $C_j = C_j - Z_j$

$\downarrow$
TOP

C_B	C_j BASIS	-1 x_1	3 x_2	-3 x_3	0 s_1	0 s_2	0 s_3	b	θ
0	s_1	3	-1	2	1	0	0	7	
0	s_2	-2	-4	0	0	1	0	12	
0	s_3	-4	3	8	0	0	1	10	
$Z_j = \sum C_B a_{ij}$		0	0	0	0	0	0	0	
$C_j = C_j - Z_j$		-1	3	-3	0	0	0		

$\uparrow$ LARGEST +ve C_j value

Observe that the second column (column corresponding to x_2) has the largest free value of $G_j = c_j - z_j$ page-(10)

∴ This column is considered as KEY COLUMN & x_2 is known as INCOMING VARIABLE

Now, we have to find the OUTGOING VARIABLE by considering the last column θ .

The ' θ ' column can be obtained by dividing the θ column entries from the corresponding elements of KEY COLUMN.

cost of x_2 is 3

C_B	C_j BASIS	x_1	x_2	x_3	s_1	s_2	s_3	b	θ
0	s_1	3	-1	2	1	0	0	7	$7/-1 = -7$
0	s_2	-2	-4	0	0	1	0	12	$12/-4 = -3$
0	s_3	-4	3	8	0	0	1	10	$10/3 = 3.33 \leftarrow$
$Z_j = \sum C_B a_{ij}$		0	0	0	0	0	0	0	
$C_j = C_j - Z_j$		-1	3	-3	0	0	0		

KEY ELEMENT

minimum
POSITIVE
RATIO

∴ outgoing
variable is

s_3

In next table, we drop s_3 &
introduce x_2 with its associated
value 3 under C_B column.

PAGE (19)

Now, convert the key element to unity using the operation

$$R_3 \rightarrow \frac{R_3}{3}$$

$$R_3 \rightarrow \frac{R_3}{3}$$

[illegible]

OPERATIONS

$$R_2 \rightarrow R_2 + 4 \text{ new } R_3$$

[illegible]

NOV compute $Z_j = \sum C_B a_{ij}$ & $C_j = C_j - Z_j$ PAG (14)

C_B	C_j BASIS	-1 x_1	3 x_2	-3 x_3	0 s_1	0 s_2	0 s_3	b	θ
0	s_1	$5/3$	0	$14/3$	1	0	$1/3$	$31/3$	
0	s_2	$-22/3$	0	$32/3$	0	1	$4/3$	$76/3$	
3	x_2	$-4/3$	1	$8/3$	0	0	$1/3$	$10/3$	
$Z_j = \sum C_B a_{ij}$		-4	3	8	0	0	1	10	
$C_j = C_j - Z_j$		3	0	-11	0	0	-1		

↑ LARGEST the value $\therefore x_1$ is the INCOMING VARIABLE

The first element of Z_j is computed as follow

$$0 \cdot \left(\frac{5}{3}\right) + 0 \cdot \left(\frac{-22}{3}\right) + 3 \cdot \left(\frac{-4}{3}\right) \quad \left\{ \begin{array}{l} \text{observe } C_B \text{ column} \\ \text{1st column} \end{array} \right\}$$

second element: $0 \cdot (0) + 0 \cdot (0) + 3 \cdot (1) = 3 \quad \left\{ \begin{array}{l} C_B \text{ column} \\ \text{second column} \end{array} \right\}$

Observe that the first column (column corresponding to x_1) has the largest possible value of $c_j - z_j$. This column is considered as KEY COLUMN & x_1 is known as INCOMING VARIABLE

Now, we have to find the OUTGOING VARIABLE by considering the last column θ .

The θ column can be obtained by DIVIDING the b column entries from the corresponding elements of KEY COLUMN

C_B	C_j BASIS	-1 x_1	3 x_2	-2 x_3	0 s_1	0 s_2	0 s_3	b	θ
0	s_1	$5/3$	0	$14/3$	1	0	$1/3$	$31/3$	$\frac{31/3}{5/3} = \frac{31}{5} \leftarrow$
0	s_2	$-22/3$	0	$32/3$	0	1	$4/3$	$76/3$	$\frac{(76/3)}{(22/3)} = -\frac{76}{22}$
3	x_2	$-4/3$	1	$8/3$	0	0	$1/3$	$10/3$	$\frac{10/3}{-4/3} = -\frac{5}{2}$
Z_j		-4	3	8	0	0	1	10	
$C_j - Z_j$		3	0	-11	0	0	-1		

As $C_j - Z_j$ contains +ve value,
We have to continue the procedure.

$\frac{31}{5}$ is the
MINIMUM
POSITIVE RATIO
∴ outgoing variable is
 s_1

In next table, we drop s_1 &

introduce x_1 as incoming variable ~~in the column~~

with its associated value $\boxed{-1}$ under C_B column

from the table, observe that $\boxed{\frac{5}{3}}$ is the **KEY ELEMENT**

convert the key element to unity using

the operation $R_1 \rightarrow \frac{R_1}{(\frac{5}{3})}$

C_B	C_j BASIS	-1 x_1	3 x_2	-3 x_3	0 s_1	0 s_2	0 s_3	b	θ
-1	x_1	1	0	$14/5$	$3/5$	0	$1/5$	$31/5$	
0	s_2								
3	x_2								

NOW, convert the elements of **key column** to zero using the elementary row operations

$$R_2 \rightarrow R_2 + \frac{92}{3} \text{ new } R_1 \quad R_3 \rightarrow R_3 + \frac{4}{3} \text{ new } R_1$$

Also construct $Z_j = \sum C_B a_{ij}$ & $C_j = c_j - Z_j$ row

C_B	C_j BASIS	x_1	x_2	x_3	s_1	s_2	s_3	b	θ
-1	x_1	1	0	14/5	3/5	0	1/5	31/5	
0	s_2	0	0	156/5	22/5	1	14/5	354/5	
3	x_2	0	1	32/5	4/5	0	3/5	58/5	
Z_j		-1	3	82/5	9/5	0	8/5	143/5	
$C_j = c_j - Z_j$		0	0	-92/5	-9/5	0	-8/5		

clearly $C_j - Z_j \leq 0$
for all values

Observe the last row.

As each $C_j - Z_j$ value is ≤ 0 , we can stop the procedure.

From the **Basis** column & **b** column, we have

Basis	b
x_1	$31/5$
s_2	$354/5$
x_2	$58/5$
Z_j	$143/5$

∴ Optimal solution is

$$x_1 = 31/5$$

$$x_2 = 58/5$$

$$x_3 = 0 \text{ (NON BASIC)}$$

$$\& Z_{\max} = 143/5$$

$$\begin{aligned} \therefore Z_{\min} &= -Z_{\max} \\ &= -143/5 \end{aligned}$$